

EMCH 260

SOLID MECHANICS

Ming Hu

Associate Professor
Department of Mechanical Engineering
University of South Carolina
E-Mail: hu@sc.edu

Axial Load

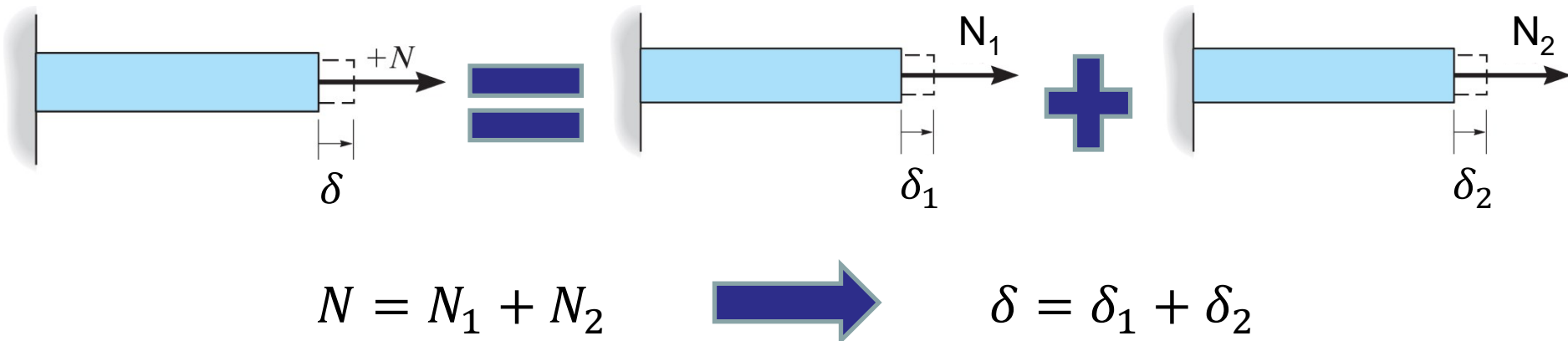
4

Chapter 4.3

- ✓ Apply the principle of superposition for total effect of different loading cases
- ✓ Use 'force method' of analysis to solve the statically indeterminate axially loaded problems

PRINCIPLE OF SUPERPOSITION

- The principle of superposition is often used to determine the stress/displacement at a point in a member when the member is subjected to complicated loading.
- After subdividing the load into components, the **principle of superposition** states that the resultant stress or displacement at the point can be determined by first finding the stress or displacement caused by each component load acting **separately** on the member.
- Resultant stress/displacement determined **algebraically** by adding the contributions of each component.



PRINCIPLE OF SUPERPOSITION, CONT'D

Conditions

1. The loading must be *linearly* related to the stress or displacement that is to be determined. E.g. in previous example, $\delta = \frac{NL}{AE}$.
2. The loading must *not significantly* change the original geometry or configuration of the member. E.g. deformation in elastic range.

When to ignore deformations?

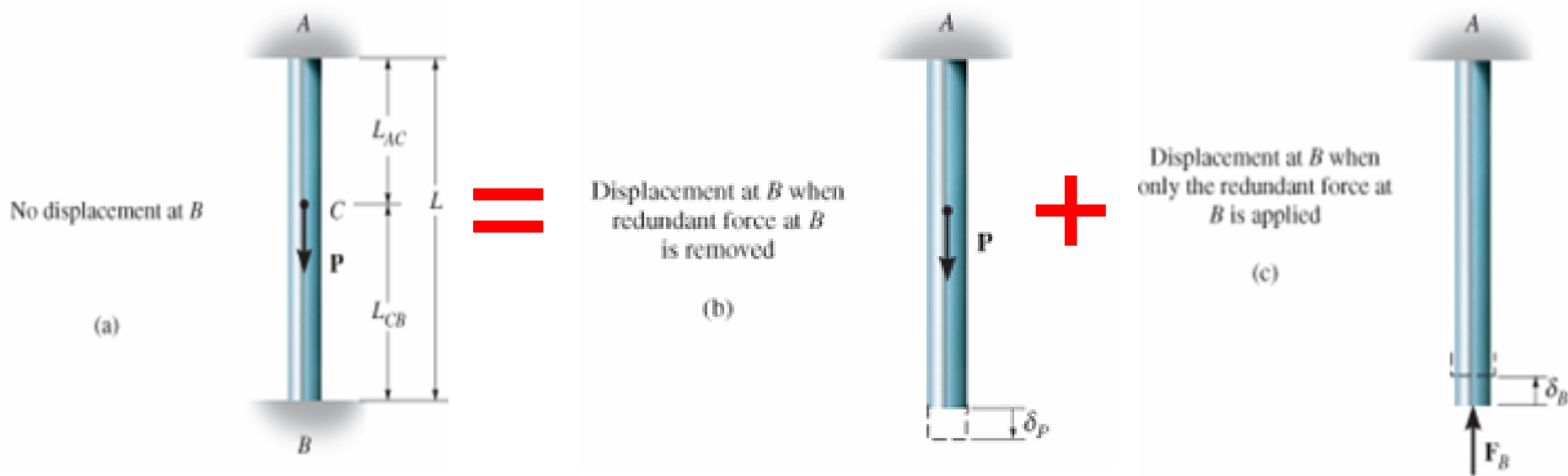
- Most loaded members will produce deformations so small that change in position and direction of loading will be insignificant and can be *neglected*

THE FORCE METHOD OF ANALYSIS FOR AXIALLY LOADED MEMBERS

- Used to also solve statically indeterminate problems by using superposition of the forces acting on the free-body diagram
- First, choose any one of the two supports as “*redundant*” and remove its effect on the bar
- Thus, the bar becomes statically determinate
- Second, *apply principle of superposition* and solve the equations simultaneously

THE FORCE METHOD (CONT'D)

- From free-body diagram, we can determine the reaction at A



PROCEDURE FOR ANALYSIS

Step 1: Compatibility

- Choose one of the supports as redundant and write the equation of compatibility.
- Known displacement at redundant support (usually zero), equated to displacement at support caused *only* by external loads acting on the member *plus* the displacement at the support caused *only* by the redundant reaction acting on the member.

Step 2: Load-Displacement

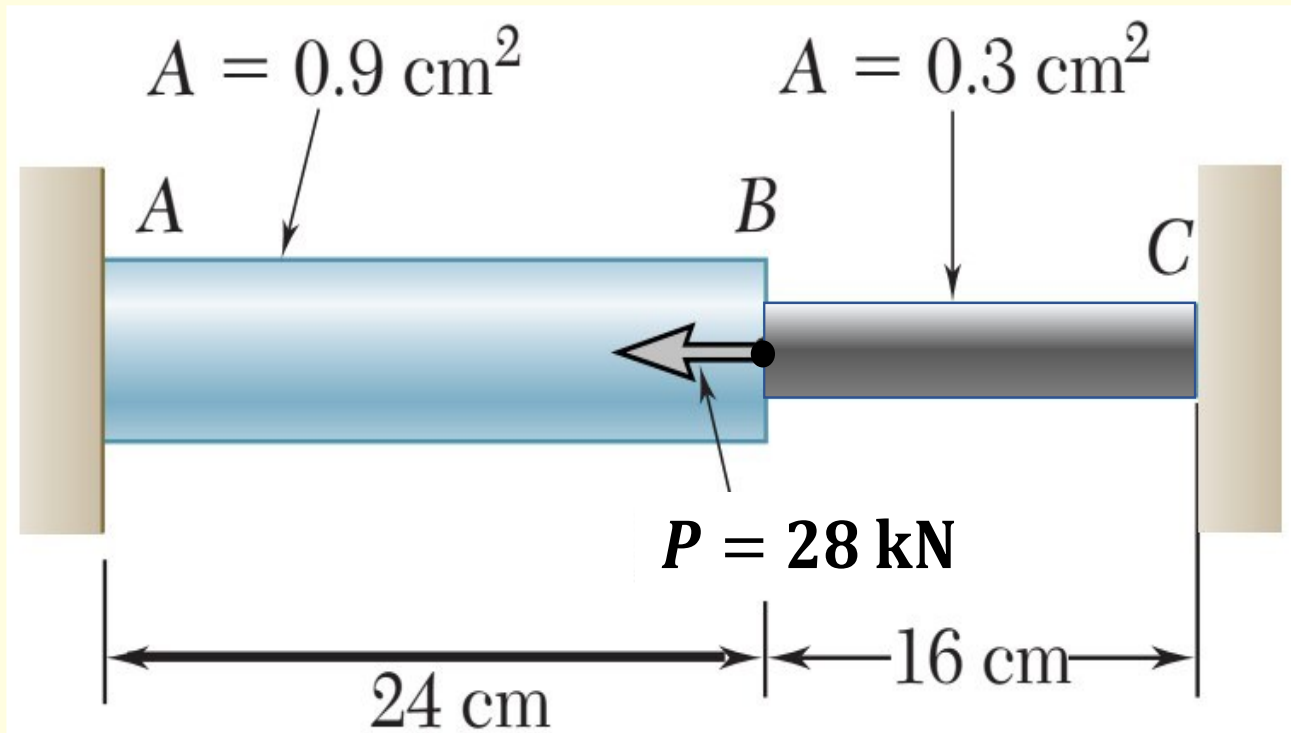
- Express external load and redundant displacements in terms of the loadings using load-displacement relationship
- Use compatibility equation to solve for magnitude of redundant force

Step 3: Equilibrium

- Draw a free-body diagram and write appropriate equations of equilibrium for member using calculated result for redundant force.
- Solve the equations for other reactions

IN-CLASS EXAMPLE 1

The cylindrical bar composed of two parts (AB and BC) is fixed at ends A and C . The two parts are made of different materials with different cross-sections. An external load $P = 28 \text{ kN}$ is applied at point B . The cross-sectional area of AB and BC part is 0.9 cm^2 and 0.3 cm^2 , respectively. The modulus of elasticity of the AB and BC part is 40 GPa and 200 GPa , respectively. Determine the force reactions at ends A and C .



IN-CLASS EXAMPLE 1, cont'd

Solutions

This is an indeterminate axially loaded member.

Step 1: Equilibrium

Assuming the resultant force reactions at the two ends A and C as shown in the figure, for the entire bar, the Free-Body-Diagram (force equation of equilibrium) gives:

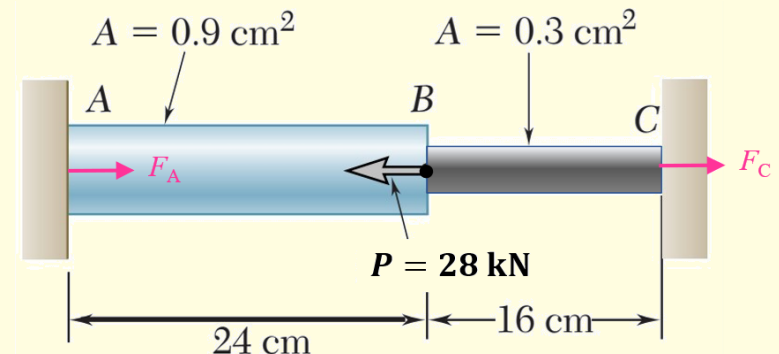
$$\sum F_x = 0 \rightarrow$$

$$F_A + F_C = P = 28 \text{ kN} \quad (1)$$

Step 2: Compatibility

The total displacement of the entire bar (from A to C) should be zero, i.e.

$$\delta_{A/C} = 0 \rightarrow \delta_{AB} + \delta_{BC} = 0 \quad (2)$$



IN-CLASS EXAMPLE 1, cont'd

Solutions

Step 3: Load-Displacement Relations

The elongation / compression of the bar can be obtained by the force (related to stress) and modulus of elasticity:

$$\delta_{AB} = - \frac{F_A L_{AB}}{E_{AB} A_{AB}}$$

$$\delta_{BC} = + \frac{F_C L_{BC}}{E_{BC} A_{BC}}$$

Substituting them into Eq. (2), we get

$$\frac{F_A L_{AB}}{E_{AB} A_{AB}} = \frac{F_C L_{BC}}{E_{BC} A_{BC}}$$

$$\rightarrow \frac{F_A \times 24}{40 \times 10^9 \times 0.9} = \frac{F_C \times 16}{200 \times 10^9 \times 0.3}$$

$$\rightarrow F_A = 0.4 F_C \quad (3)$$

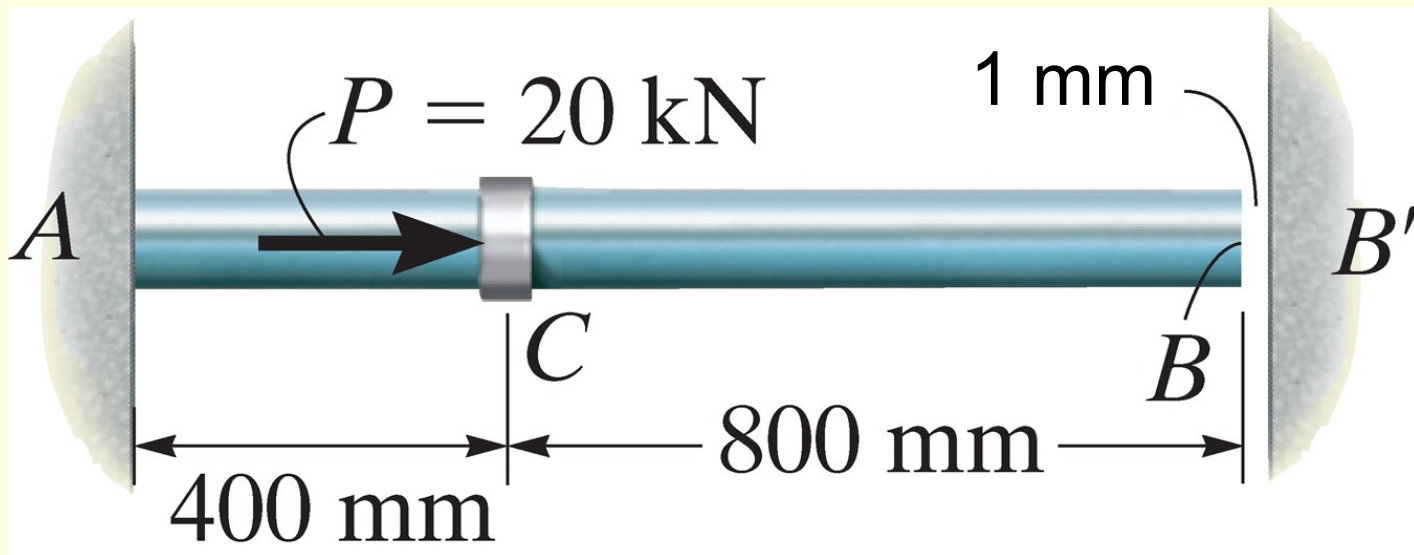
Combining Eqs. (1) and (3) we obtain:

$$F_A = 8 \text{ kN}, \quad F_C = 20 \text{ kN}$$

Note: Both final reaction forces are positive, indicating that our presumed direction of the reaction forces is correct, i.e. the *A* end is compressive force and the *C* end is tensile force.

IN-CLASS EXAMPLE 2

The steel rod has a diameter of 5 mm. It is fixed to the wall at A. Before it is loaded, there is a gap of 1 mm between the wall at B' and the rod. Determine the reactions at A and B' if it is subjected to an axial force $P = 20$ kN. Neglect the size of the collar at C. Take Young's modulus of steel as 200 GPa.



IN-CLASS EXAMPLE 2, cont'd

Solutions

Compatibility

Consider support at B' as redundant. Use principle of superposition, we have

$$0.001 \text{ m} = \delta_P - \delta_B$$

Equation 1

Deflections δ_P and δ_B are determined by

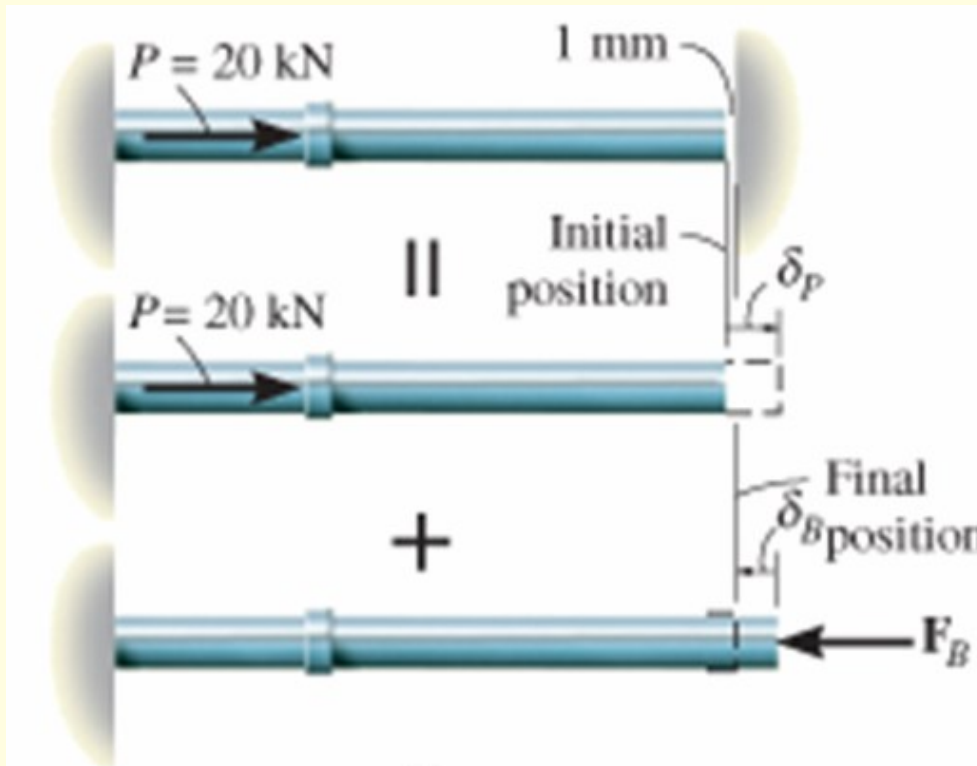
$$\delta_P = \frac{PL_{AC}}{AE} = \dots\dots = 0.002037 \text{ m}$$

$$\delta_B = \frac{F_B L_{AB}}{AE} = \dots\dots = 0.3056 \times 10^{-6} F_B$$

Equation 2

Substituting Eq. (2) into (1), we get

$$F_B = 3.40 \text{ kN}$$



IN-CLASS EXAMPLE 2, cont'd

Solutions

Equilibrium

From free-body diagram of the entire rod, we have

$$\sum F_x = 0$$

$$-F_A + 20 \text{ kN} - 3.4 \text{ kN} = 0$$

$$F_A = 16.6 \text{ kN} \quad \textbf{(Ans.)}$$

The positive sign indicates the assumed direction of force F_A is correct.



Try to solve this problem with the method taught in Ch. 4.2

IN-CLASS EXAMPLE 2, cont'd

Another Solution:

If there is no wall B' , the CB part will be moving to the right by a length:

$$\delta_P = \frac{PL_{AC}}{AE} = \dots\dots = 0.002037 \text{ m} > 1 \text{ mm (gap)}$$

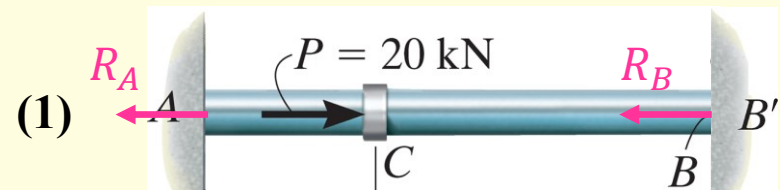
This means the end B will be in contact with wall B' when the wall exists.

Step 1: Equilibrium

Assuming the resultant reactions at the two ends A and B as shown in the figure, for the entire bar, the free-body diagram (force equation of equilibrium) gives:

$$\sum F_x = 0 \rightarrow$$

$$R_A + R_B = P$$



Step 2: Compatibility Condition

The total strain or displacement of the entire bar (from A to B) should equal to the gap:

$$\delta_{A/B} = 0 \rightarrow$$

$$\delta_{AC} + \delta_{CB} = 1 \text{ mm}$$

(2)

IN-CLASS EXAMPLE 2, cont'd

Another Solution:

Step 3: Load-Displacement Relations

The elongation / compression of the bar can be obtained by the force (related to stress) and modulus of elasticity (refer to lecture ppt Ch.4.1 page 7):

$$\delta_{AC} = + \frac{R_A L_{AC}}{EA}$$

$$\delta_{CB} = - \frac{R_B L_{CB}}{EA}$$

Substituting them into Eq. (2), we get

$$\frac{R_A L_{AC}}{EA} - \frac{R_B L_{CB}}{EA} = 1 \text{ mm}$$

→

$$R_A - 2R_B = 9.8 \text{ kN} \quad (3)$$

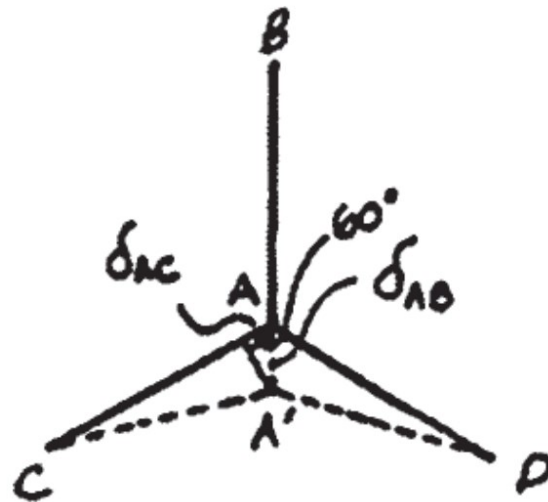
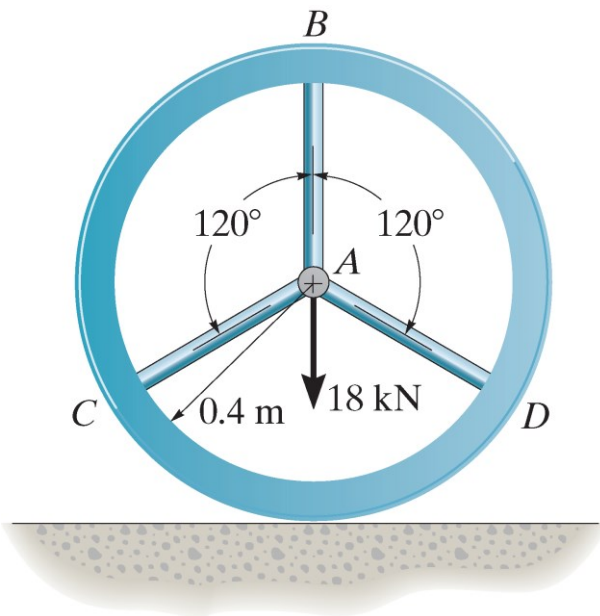
Combining Eqs. (1) and (3), we obtain:

$$R_A = 16.6 \text{ kN} , \quad R_B = 3.4 \text{ kN} \quad (\text{Ans.})$$

COMPATIBILITY OF SOME EXAMPLES

For indeterminate uniaxial loaded problems, you should always think about displacement or geometry conditions at first.

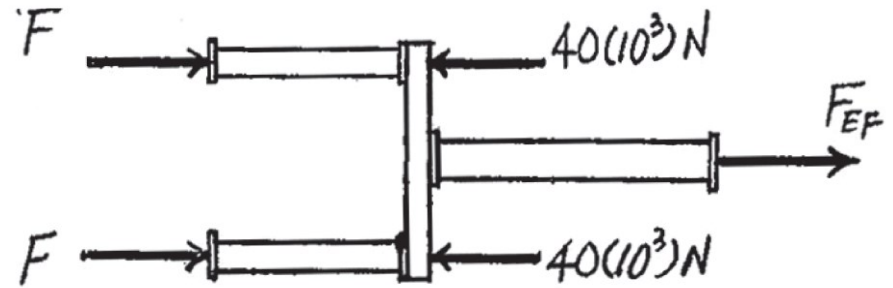
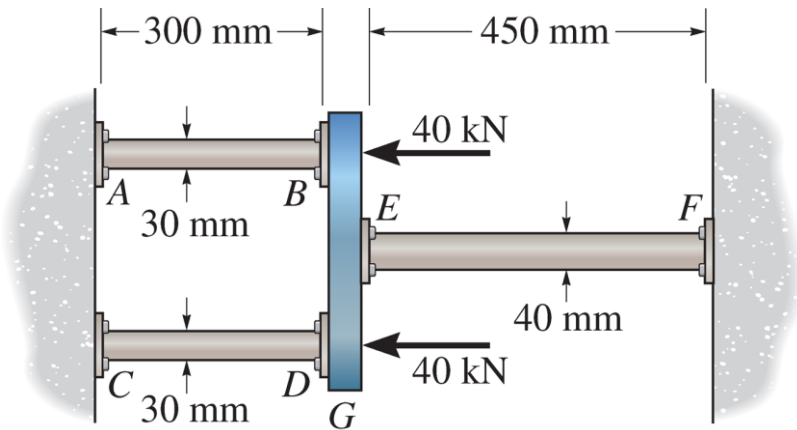
Example 1:



$$\text{Compatibility: } \delta_{AC} = \delta_{AB} \cos 60^\circ$$

COMPATIBILITY OF SOME EXAMPLES

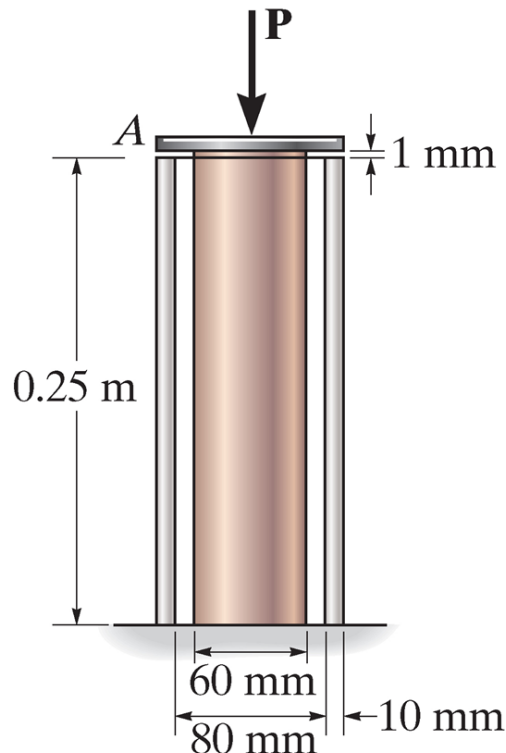
Example 2:



Compatibility: $\delta_{A/F} = \delta_{AB} + \delta_{EF} = 0$

COMPATIBILITY OF SOME EXAMPLES

Example 3:

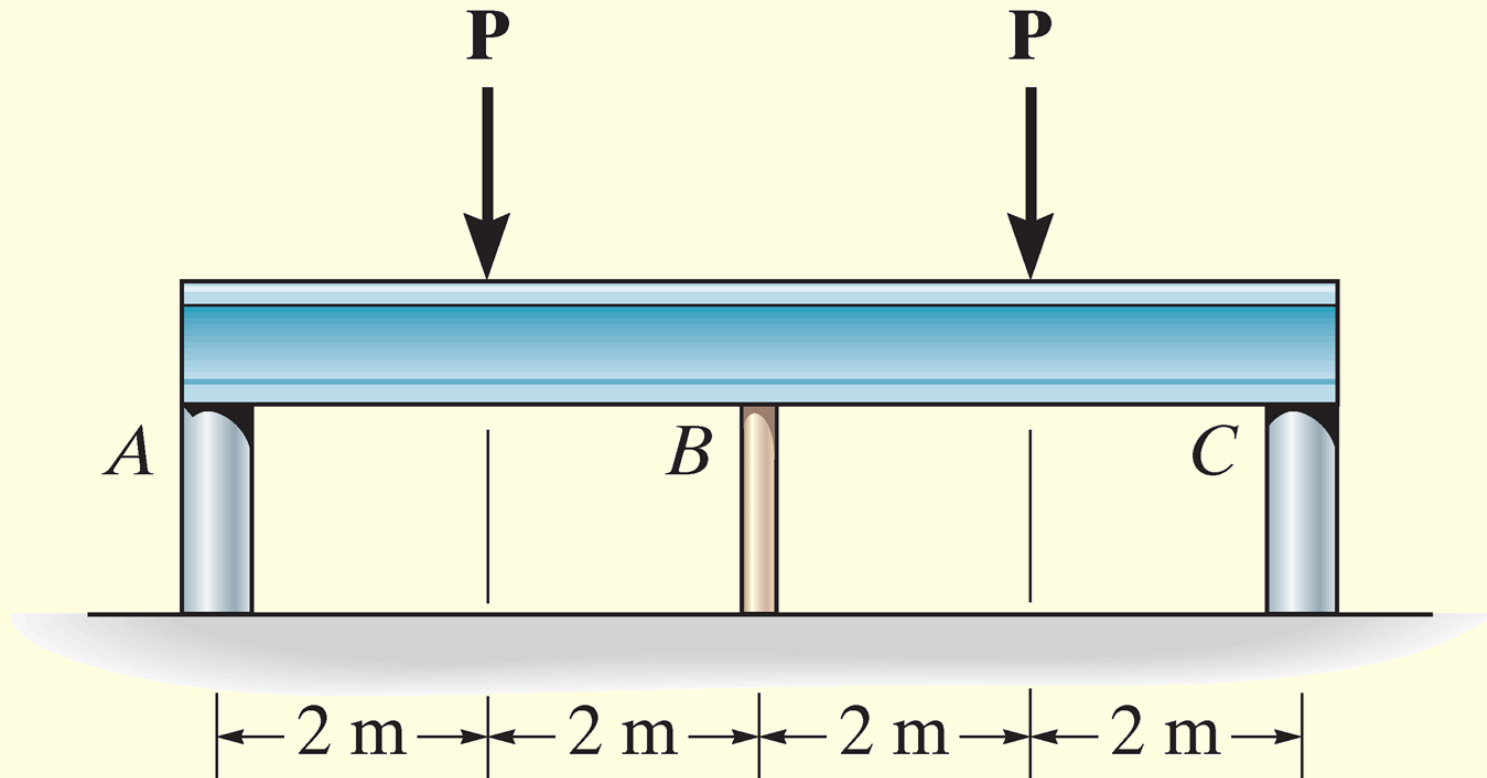


The support consists of a solid red brass copper post surrounded by a stainless steel tube. **Before the load is applied the gap between these two parts is 1 mm.** Given the dimensions shown, determine the greatest axial load that can be applied to the rigid cap *A* without causing yielding of the steel tube. Assume the elastic deformation of brass and the Young's modulus for brass is 105 GPa. The Yield strength and Young's modulus for steel is 207 MPa and 193 GPa, respectively.

$$\text{Compatibility: } \delta_{\text{steel}} = \delta_{\text{brass}} - 0.001$$

IN-CLASS EXAMPLE 3

The rigid beam is supported by the three posts A , B , and C of equal length. Posts A and C have a diameter of 75 mm and are made of alumina for which $E = 70$ GPa and $\sigma_Y = 20$ MPa. Post B has a diameter of 20 mm and is made of brass for which $E' = 100$ GPa and $\sigma_Y' = 590$ MPa. Determine the smallest magnitude of P so that only rods A and C yield.



IN-CLASS EXAMPLE 3, cont'd

Solutions

Step 1: Equilibrium

Assume the resultant forces in the posts and apply free body diagram to the rigid beam:

$$\sum M_B = 0 \quad \rightarrow \quad F_A = F_C = F_{al} \quad (1)$$

$$\sum F_y = 0 \quad \rightarrow \quad 2F_{al} + F_{br} = 2P \quad (2)$$

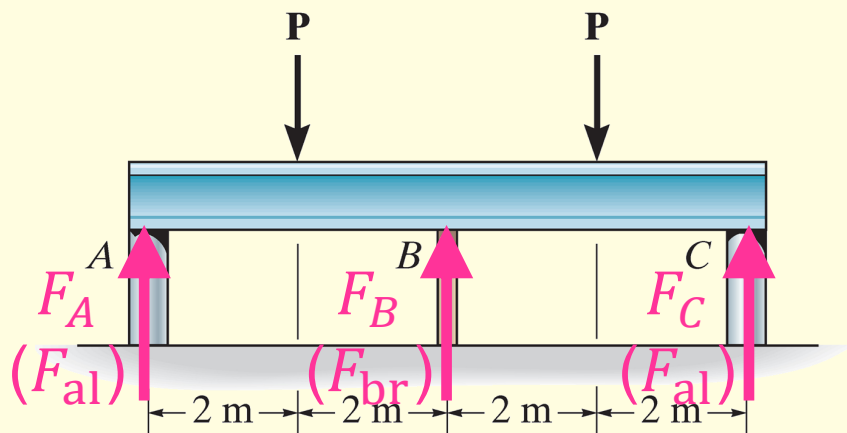
Step 2: Compatibility Condition

Referring to the geometry of the deformation of the three posts (A and C should have the same deformation, noted as δ_{al}):

$$\delta_{al} = \delta_{br} \quad (3)$$

$$\delta_{al} = \frac{F_{al}L}{E_{al}A_{al}}$$

$$\delta_{br} = \frac{F_{br}L}{E_{br}A_{br}}$$



IN-CLASS EXAMPLE 3, cont'd

Solutions, cont'd

Substituting these equations into Eq. (3) we get:

$$\frac{F_{al}L}{E_{al}A_{al}} = \frac{F_{br}L}{E_{br}A_{br}}$$

$$\frac{F_{al}}{70 \times 10^9 \times \frac{\pi}{4} \times 0.075^2} = \frac{F_{br}}{100 \times 10^9 \times \frac{\pi}{4} \times 0.02^2}$$

$$F_{br} = 0.1016F_{al} \quad (4)$$

Substituting Eq. (4) into Eq. (2) yields:

$$F_{al} = 0.95166P \quad (5)$$

Since post A and C will yield,

$$F_{al} = (\sigma_Y)_{al} \cdot A_{al} = 20 \times 10^6 \times \frac{\pi}{4} \times 0.075^2 = 88.36 \text{ kN}$$

Substituting into Eq. (5) we get

$$P = 92.85 \text{ kN} \quad (\text{Ans.})$$

IN-CLASS EXAMPLE 3, cont'd

Solutions, cont'd

Make sure the post B will not yield:

Combining the force P and Eqs. (4) & (5) we have

$$F_{\text{br}} = 0.1016F_{\text{al}} = 0.09669P = 8.9775 \text{ kN}$$

Then, the compressive stress in post B is

$$\sigma_B = \frac{F_{\text{br}}}{A_{\text{br}}} = \frac{8977.5}{\frac{\pi}{4} \times 0.02^2} = 28.58 \text{ MPa} < (\sigma_Y)_{\text{br}}$$

O.K.

OUT-OF-CLASS READING

- **Textbook, pp. 140, 147 – 148**